

AARON DAVIDGE

Software Engineer · Cloud · Backend · AI Systems

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SUMMARY

Self-taught builder who ships to production. Computer science foundation with roughly 110 credit hours of CS and mathematics coursework, plus working command of cloud deployment: containerized FastAPI services on Google Cloud Run behind an Application Load Balancer, fronting Gemini, with Model Armor screening traffic in both directions. I own builds end to end, from requirements through production deploy.

PROJECTS

AARON.AI — Conversational Agent Backend | Cloud Run · Docker · FastAPI · Load Balancer · Model Armor · Gemini | aarondavidge.com/assistant.html

- Built and deployed a containerized FastAPI agent on Cloud Run, fronted by a Google Cloud Application Load Balancer at api.aarondavidge.com, serving the public concierge on my site.
- Routed all traffic through Model Armor, screening prompts and responses for injection and unsafe content before they reach the model or the user.
- Designed the agent in Agent Studio and packaged it in Docker for reproducible deploys; ships with a visible "AI · can be wrong" disclosure and a human-escalation path.
- Deliberately scoped it to conversational Q&A it can answer reliably rather than shipping ungrounded retrieval — RAG and tool calling are the next milestones.

SAM.gov Opportunities MCP Server — Natural-Language Interface to Federal Procurement Data | Model Context Protocol · Python · REST API integration · Antigravity CLI | *working prototype*

- Built a Model Context Protocol (MCP) server exposing the federal SAM.gov Opportunities API as callable tools, letting an agent client retrieve live contract solicitations.
- Wired the Antigravity CLI as the client so plain-language questions resolve into parameterized queries against the live API — no hand-built search syntax.

Zero-Day Intelligence Pipeline — Automated Ingestion Service | Python · Web scraping · Scheduled ingestion | github.com/44r0nd4vidg3

- Automated pipeline that ingests Google Project Zero's root-cause analyses of in-the-wild zero-days and republishes them as structured, queryable records — parsing inconsistent source formatting and re-running on a schedule to pick up new disclosures without duplicating entries.

aarondavidge.com — Multi-Subdomain Web Platform & Admin Dashboard | React · Firebase Hosting & Authentication · GitHub API · Tailwind CSS · GitHub Pages · GitHub Actions | aarondavidge.com

- Designed and shipped the hub tying together the home, blog, store, and assistant surfaces with full SEO and Open Graph metadata, hand-built without a framework to keep the bundle small.
- Split the platform across two hosts: static assets and blog content version-controlled in a GitHub repository, with the React admin SPA (built in Google AI Studio) deployed separately on Firebase Hosting behind Firebase Authentication.
- Wired the dashboard to publish new posts directly into the GitHub repository through the GitHub API, so content changes flow through version control rather than a separate CMS.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C++, Bash, Assembly, HTML/CSS, SQL (learning)
- **Cloud & Infrastructure:** Google Cloud (Cloud Run, Application Load Balancing, Model Garden), Firebase (Hosting, Authentication), Docker, containerized deployment, GitHub Actions, GitHub Pages, GitHub API, Git
- **Backend:** FastAPI, REST API design, Express, MongoDB
- **AI & Agents:** Model Context Protocol (MCP), tool calling, Gemini, Gemini Enterprise Agent Platform, Gemini Enterprise for CX, Dialogflow CX, Agent Studio, Google ADK, Google AI Studio (incl. app build), Model Armor, Antigravity, prompt and context engineering
- **Frontend:** React, Redux, Firebase Authentication / Google sign-in, Tailwind CSS, vanilla JavaScript, semantic HTML, responsive layout, SEO and Open Graph
- **Security practice:** Secure-by-default design and secrets hygiene, system decoupling, input sanitization and injection screening, network segmentation
- **Currently building with:** RAG and vector search, multi-agent orchestration, agent evaluation, PostgreSQL, Stripe, Flutter

EDUCATION

B.S. Computer Science coursework — approximately 110 credit hours completed at Southeastern Louisiana University and Louisiana State University. Not currently enrolled; open to completing remaining requirements.

- **Computer science:** Algorithms I & II (Java), Data Structures (C++), Object-Oriented Design, Theory of Computation, Assembly, Enterprise Development, Software Engineering (React, Redux, Express, MongoDB), Systems Administration (Bash), Data Science (Python).
- **Mathematics:** Calculus I, II & III, probability and statistics, differential equations.
- **Applied coursework labs:** Conversational agent development in Dialogflow CX and Gemini Enterprise for CX — built a donor-eligibility screening agent, and a retrieval-grounded agent answering over unstructured documents via a CX datastore.